Job, Batch, and Flow Production Methods

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Job Production Overview

- Definition: Producing customized products per client specifications.
- Examples: Furniture, custom machinery, art.
- Characteristics:
- - High flexibility
- - Skilled labor required
- - Longer production time

Job Production

- Scenario:
- - A custom machine costs $100/hour to produce.
- - Labor hours required: 250.
- Question: What is the production cost?
- Solution:
- Cost = Labor Rate × Hours = 100 × 250 = $25,000

Job Production Example



Batch Production Overview

- Definition: Producing a set quantity of identical items in batches.
- Examples: Clothing, bakery products.
- Characteristics:
- - Medium flexibility
- - Economies of scale
- - Idle time between batches

Batch Production

- Scenario:
- - Production setup cost: $2,000
- - Cost per unit: $50
- - Batch size: 500 units
- Question: What is the total production cost per unit?
- Solution:
- Total Cost = Setup Cost + (Cost per Unit × Batch Size)
- Total Cost per Unit = (2000 + (50 × 500)) / 500 = $54

Batch Production Example



Flow Production Overview

- Definition: Continuous production of standardized products.
- Examples: Cars.
- Characteristics:
- - Low flexibility
- - High efficiency
- - Significant initial investment

Flow Production

- Scenario:
- - Production rate: 100 units/hour
- - Operating hours: 8/day
- - Operating days: 300/year
- Question: What is the annual production?
- Solution:
- Annual Production = Rate × Hours/Day × Days/Year = 100 × 8 × 300 = 240,000 units

Flow Production Example



Comparing Production Methods

- Characteristic Comparison:
- | Flexibility | Job: High | Batch: Medium | Flow: Low |
- | Efficiency | Job: Low | Batch: Medium | Flow: High |
- | Setup Cost | Job: Low | Batch: Medium | Flow: High |
- Applications:
- - Job: Custom products
- - Batch: Medium-volume goods
- - Flow: Mass production

Case Study: Combined Methods

- Scenario:
- A company manufactures 3 types of products:
- - Product A: 100 custom units/year (Job).
- - Product B: 10,000 units/year in batches of 500.
- - Product C: 500,000 units/year in flow.
- Task: Calculate total costs for each method and combined.

Solution: Case Study

- Calculation:
- - Job: 100 × $1,000 = $100,000
- - Batch: 10,000 × $50 + (10,000 / 500) × $10,000 = $510,000
- - Flow: 500,000 × $10 = $5,000,000
- Total: $100,000 + $510,000 + $5,000,000 = $5,610,000

Key Takeaways

- Job Production: Best for customization.
- Batch Production: Balances flexibility and cost.
- Flow Production: Best for efficiency and large volumes.

Process-Focussed System



Process-Focused System



Job production

Merits	Demerits
▪ Quality is very high as workers can concentrate on one product ▪ Work variety is high. A person gets lot of experience. Thus, workers are motivated ▪ Products can be customized with detailed specifications ▪ Skilled labor ensures better quality and value addition	▪ High labor cost as workers must be skilled in a number of areas ▪ Production can be slow due to difficulty of planning ▪ A wide range of specialist tools, persons and machine may be needed ▪ Expensive and inefficient ▪ Unsuitable for higher-batch production, lacks economy of scale

Batch production

Merits	Demerits
▪ Workers can be specialized in one process or in only a few processes ▪ Per Unit costs are lower and output is higher ▪ Different orders can be met for multiple customers in batches ▪ There is good scope of automation ▪ Some materials can be reused between batches, thereby reducing waste	▪ Less motivation as workers will be more specialized and may find it monotonous ▪ More planning and coordination needed ▪ More money is tied up in the production process. Set up costs may be high ▪ Small batches can still have high costs. No proper economy of scale ▪ Delayed defect detection may result in the entire batch being discarded

Flow production

Merits	Demerits
❖ Low per unit cost due to economy of scale ❖ Rate of production is high ❖ Modern technology allows full automation ❖ Production speed can be monitored according to demand ❖ Efficient production process ensures minimum downtime	❖ Set up costs are very high ❖ Tasks can be very repetitive and monotonous; hence, low worker motivation ❖ Breakdown in production can be very expensive for the business. Hence, proper maintenance should be there. ❖ High storage requirement for raw materials and finished products, if no proper inventory management is there.

Mass customization

- **Mass customization** is a manufacturing strategy that combines the efficiency of mass production with the flexibility of customized products. It allows companies to produce goods and services tailored to individual customer preferences while still benefiting from the cost efficiencies associated with large-scale production.

Key features of mass customization

- **Personalization**: Products or services are adapted to meet specific customer needs or preferences.
- **Efficiency**: Utilizes standardized components and modular design to keep costs low.
- **Technology Integration**: Relies heavily on advanced technologies like CAD (Computer-Aided Design), CAM (Computer-Aided Manufacturing), and digital twins for designing, producing, and delivering customized products.
- **Customer Involvement**: Customers are often involved in the design process, such as selecting features or configurations.

Examples of mass customization

- **Nike By You**: Nike allows customers to design their shoes by choosing colors, materials, and styles.
- **Dell Computers**: Customers can configure the specifications of their laptops or desktops, including memory, storage, and processor.
- **Customized Automobiles**: Car manufacturers like BMW let buyers choose colors, interiors, and additional features.
- **Online Apparel Brands**: Companies like Threadless or Indochino offer customized clothing designs or tailored suits.

Benefits of mass customization

- **Improved Customer Satisfaction**: Tailored products better meet customer needs.
- **Competitive Advantage**: Differentiates brands in crowded markets.
- **Optimized Inventory**: Reduced risk of overstocking unwanted items.

Challenges of Mass Customization

- **Complexity in Operations**: Managing diverse customer demands while maintaining efficiency.
- **Higher Initial Investment**: Requires advanced technology and flexible manufacturing systems.
- **Supply Chain Management**: Coordination across the supply chain becomes more complex.

Types of Mass Customization

- **Collaborative Customization**: Customers work with companies to design a product that meets their specific needs.
- **Adaptive Customization**: A standard product is designed to be adjusted by the user (e.g., adjustable office chairs).
- **Transparent Customization**: Products are tailored for the customer without explicit requests
- **Cosmetic Customization**: Standard products are presented differently to various customers (e.g., personalized packaging).

Legacy of Fordism

- - Concept of mass production with lower prices and higher wages
- - Spread worldwide, influencing industrial practices
- - Innovations:
- - Higher productivity
- - Improved living standards